Decision Tree Activity

Ol Outlook	T Temp	H Humidity Hot	W Windy	R Rain Prediction
Sunny	Hot	High	Weak	No
Sunny	Mild	High	Strong	No
Cloudy	Hot	High	Weak	Yes
Rainy	Mild	High	Weak	Yes
Rainy	Cool	Normal	Weak	Yes
Rainy	Cool	Normal	Strong	No
Cloudy	Mild	Normal	Strong	Yes
Sunny	Cool	Normal	Weak	No

Entropy of Prime Attribute : Rain

$$= -\frac{4}{8}\log\frac{4}{8} - \frac{4}{8}\log\frac{4}{8}$$

$$= -(-0.150) - (-0.150)$$

$$= 0.301$$

Entropy of non-Prime attribute

Info gain of outlook

	Yes	No	Entropy
Sunny	0	3	0
Cloudy	2	0	0
Rainy	2	1	-0.037

∴ Info gain = Prime Entropy - Non Prime entropy

= 0.301 - (-0.037)

= 0.075

Info gain of Temperature

	Yes	No	Entropy
Hot	1	1	0
Mild	2	1	-0.037
Cool	1	2	-0.037

Ent = 0 + (-0.037) + (-0.037)

Ent = -0.074

Info gain = 0.301 - 0.074 = 0.227

Info gain of Humidity

	Yes	No	Entropy
High	2	2	0
Normal	2	2	0

Info gain = 0.301 − 0

= 0.301

Info gain of Windy

	Yes	No	Entropy
Weak	3	2	−0.0092
Strong	1	0	0.037

Entropy = −0.0092 + 0.037

= 0.027

Info gain = 0.301 − 0.027

= 0.274

(4)

compare all	Entropy
Rain	0.301
outlook	0.075
Temperature	0.227
Humidity	0.301
Windy	0.274

lowest is outlook So Root node is outlook

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23122073